Cauchy-characteristic matching for the Z4c evolution system: method

z4c-CMM verified research loop[1]

[1] *generated from the mission-2 knowledge ledger (`knowledge-database/paper_z4c-CCM/`); every equation and number below is traceable to a verifier output under `results/numerical/`*

(Dated: June 12, 2026)

I. METHOD

Black equations are transcribed from Refs. [1–3]; dark blue equations constitute the new Z4c-CCM formulation. Verification labels [Vn] refer to Table I; dynamical evidence is in Table II.

A. Boundary frame and Z4c variables

$$ds^2 = -\alpha^2\, dt^2 + \gamma_{ij}\,(dx^i + \beta^i dt)(dx^j + \beta^j dt), \qquad n^\mu = \alpha^{-1}\,(1,\, -\beta^i), \tag{1}$$

$$s_i = \frac{\partial_i r}{\sqrt{\gamma^{rr}}}, \qquad s^i = \frac{\gamma^{ir}}{\sqrt{\gamma^{rr}}}, \qquad m^A = \tfrac{1}{\sqrt{2}}\big(e_2 + i\, e_3\big)^A, \tag{2}$$

$$\mathring{l}^\mu = \tfrac{1}{\sqrt{2}}\,(n^\mu + s^\mu), \qquad \mathring{k}^\mu = \tfrac{1}{\sqrt{2}}\,(n^\mu - s^\mu) \qquad [1]. \tag{3}$$

Z4c variables [1]:

$$\chi = (\det\gamma)^{-1/3}, \quad \tilde\gamma_{ij} = \chi\,\gamma_{ij}, \quad K = \hat K + 2\Theta, \quad \tilde A_{ij} = \chi\Big(K_{ij} - \tfrac{1}{3}\gamma_{ij}K\Big), \tag{4}$$

$$\partial_t\chi = \tfrac{2}{3}\,\chi\,\big[\alpha(\hat K + 2\Theta) - D_i\beta^i\big], \tag{5}$$

$$\partial_t\tilde\gamma_{ij} = -2\alpha\tilde A_{ij} + \beta^k\tilde\gamma_{ij,k} + 2\tilde\gamma_{k(i}\,\beta^k_{\ ,j)} - \tfrac{2}{3}\,\tilde\gamma_{ij}\,\beta^k{}_{,k} \qquad [1]. \tag{6}$$

B. Worldtube data map (node N1)

$$g_{00} = -\alpha^2 + \gamma_{ij}\beta^i\beta^j, \qquad g_{0i} = \gamma_{ij}\beta^j, \qquad g_{ij} = \gamma_{ij}; \qquad g^{00} = -\alpha^{-2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2}. \tag{7}$$

ADM kinematics requires $\partial_t\gamma_{ij} = -2\alpha K_{ij} + \beta^k\partial_k\gamma_{ij} + \gamma_{kj}\partial_i\beta^k + \gamma_{ik}\partial_j\beta^k$. From $\gamma_{ij} = \chi^{-1}\tilde\gamma_{ij}$,

$$\partial_t\gamma_{ij} = \chi^{-1}\partial_t\tilde\gamma_{ij} - \chi^{-2}\tilde\gamma_{ij}\,\partial_t\chi, \tag{8}$$

and inserting (5)–(6) with $K_{ij} = \chi^{-1}\big(\tilde A_{ij} + \tfrac{1}{3}\tilde\gamma_{ij}K\big)$, $\beta^k\partial_k\gamma_{ij} = \chi^{-1}\beta^k\partial_k\tilde\gamma_{ij} - \gamma_{ij}\,\beta^k\partial_k\ln\chi$,

$$\big[\partial_t\gamma_{ij}\big]_{\rm Z4c} - \big[\partial_t\gamma_{ij}\big]_{\rm ADM} = \gamma_{ij}\Big[\tfrac{2}{3}\,D_k\beta^k - \tfrac{2}{3}\,\partial_k\beta^k + \beta^k\partial_k\ln\chi\Big]. \tag{9}$$

$$\boxed{D_i\beta^i = \partial_k\beta^k - \tfrac{3}{2}\,\beta^k\,\partial_k\ln\chi} \qquad \text{[V3: closure exact; naive } D = \partial_k\beta^k \text{ leaves residual } +1\cdot\gamma_{ij}\beta^k\partial_k\ln\chi]. \tag{10}$$

C. Tetrad correspondence and Type-II boost (node N2)

Outgoing null label u with $u|_{\partial\Omega} = t$: $\partial_\mu u = (1,\, u_r,\, 0,\, 0)$,

$$g^{tt} + 2g^{tr}u_r + g^{rr}u_r^2 = 0 \quad\Longrightarrow\quad u_r = \frac{-g^{tr} - \sqrt{(g^{tr})^2 - g^{tt}g^{rr}}}{g^{rr}}, \qquad u_r\big|_{\text{flat}} = -1. \tag{11}$$

Characteristic outgoing null vector [2, 3]:

$$\hat{l}^{\,\mu} = -e^{2\hat{\beta}}\, g^{\mu\nu}\partial_\nu u. \tag{12}$$

For arbitrary $(\alpha, \beta^i, \gamma_{ij})$,

$$\mathring{l}^\mu = \frac{\sqrt{2}}{2}\left(\alpha - \gamma_{ij}\beta^i s^j\right) e^{-2\hat{\beta}}\, \hat{l}^{\,\mu} \qquad \text{[V1: exact at 12 rational points; GPU residual } 9.3\times10^{-16}\text{]}, \tag{13}$$

$$A = \left(\alpha - \gamma_{ij}\beta^i s^j\right) e^{-2\hat{\beta}}, \qquad \psi_0' = A^2\,\psi_0, \tag{14}$$

i.e. the Choice-2 boost of Ref. [3] holds verbatim on the Z4c frame (3): no evolution-system variable enters (13).

ψ_0 on the characteristic grid [2, 3]:

$$\psi_0 = \Big(\frac{r\partial_r\beta - 1}{4Kr}\Big)\Big[(1{+}K)\partial_r J - \frac{J^2\partial_r\bar{J}}{1{+}K}\Big] + \frac{J(1{+}K^2)\,\partial_r J\,\partial_r\bar{J}}{8K^3} + \frac{1}{8K}\Big[\frac{J^2\partial_r^2\bar{J}}{1{+}K} - (1{+}K)\partial_r^2 J\Big] - \frac{J\bar{J}^2(\partial_r J)^2 + J^3(\partial_r\bar{J})^2}{16K^3}. \tag{15}$$

D. Physical-mode dictionary (node N3, exact)

Electric/magnetic Weyl decomposition w.r.t. n^μ (vacuum):

$$E_{\mu\nu} = C_{\mu\alpha\nu\beta}\, n^\alpha n^\beta, \qquad B_{\mu\nu} = \tfrac{1}{2}\,\epsilon_{\mu\alpha}{}^{\gamma\delta}C_{\gamma\delta\nu\beta}\, n^\alpha n^\beta, \qquad \epsilon_{\mu\nu\rho\sigma} = \sqrt{-g}\,[\mu\nu\rho\sigma]. \tag{16}$$

EXACT identity (pure Weyl algebra, generic 10-component tensor; no perturbation, no field equations) [V2a]:

$$\left(E_{\mu\nu} + \epsilon_\mu{}^{\kappa\lambda}s_\kappa B_{\lambda\nu}\right) m^\mu m^\nu \;=\; -\,\psi_0, \qquad \left(E_{\mu\nu} - \epsilon_\mu{}^{\kappa\lambda}s_\kappa B_{\lambda\nu}\right) m^\mu m^\nu \;=\; -\,\bar{\psi}_4, \tag{17}$$

with $\psi_0 = -C_{abcd}\, l^a m^b l^c m^d$ on the frame (3); dyad completeness ($U_{AB} = U(m,m)\,\bar{m}_A\bar{m}_B + \text{c.c.}$ for tangential trace-free U) gives the tensor form of the incoming Weyl object

$$U^-_{AB} \;\equiv\; \mathrm{TT}_2\big[E + \epsilon(s)B\big]_{AB} \;=\; -\big(\psi_0\,\bar{m}_A\bar{m}_B + \bar{\psi}_0\, m_A m_B\big). \tag{18}$$

Realization from the Z4c state (Gauss–Codazzi, on-shell vacuum; γ_{ij}, K_{ij} from (4)) [V2c]:

$$E_{ij} = R_{ij}[\gamma] + K\,K_{ij} - K_{ik}K^k{}_j, \qquad B_{ij} = \epsilon_i{}^{kl}\, D_k K_{lj}\big|_{\text{sym}}, \qquad \epsilon_{ikl} = \sqrt{\det\gamma}\,[ikl]. \tag{19}$$

Nonperturbative verification (Kerr–Schild Schwarzschild, autodiff, both routes; Table I V2c): $|E^{3+1}{-}E^{4D}|$ rel. $7.7{\times}10^{-17}$, $|B^{3+1}{-}B^{4D}|$ rel. $3.9{\times}10^{-17}$, identity residual $8.5{\times}10^{-16}$. Linearized on-shell flat limit (superseded defining form, now a corollary [V2b]): for TT $h_{AB}(t, x_s)$, $\psi_0 = \frac{1}{4}(\partial_t + \partial_s)^2 h_{mm}$ and $U^-_{AB} \to -\frac{1}{4}(\partial_t + \partial_s)^2\gamma^{\text{TF}}_{AB}$, i.e. $(\partial_t + \partial_s)^2\gamma^{\text{TF}}_{AB} = 4(\psi_0\bar{m}_A\bar{m}_B + \text{c.c.})$.

E. The Z4c-CCM boundary conditions (node N6)

Ten incoming modes at the worldtube; $\mathring{X}^\mu_v = (n^\mu + \sqrt{v}\, s^\mu)/\sqrt{2}$ at sector speed v; $L \geq 1$:

$$\text{physical (2):} \qquad U^-_{AB} \mathrel{\hat{=}} -\Big(\psi_0'\,\bar{m}_A\bar{m}_B + \bar{\psi}_0'\, m_A m_B\Big), \qquad \psi_0' = A^2\,\psi_0^{\text{CCE}} \quad \text{(exact, (18)–(19))}, \tag{20}$$

$$\text{constraint (4):}\quad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}\,\Theta \hat{=} 0,\qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}\,Z_s \hat{=} 0,\qquad \big(r^2\,\mathring{l}\cdot\partial\big)^{L+1}\,Z_A \hat{=} 0,\tag{21}$$

$$\text{gauge (4):}\quad \big(r^2\,\mathring{m}\cdot\partial\big)^{L+1}\alpha \hat{=} 0,\qquad \big(r^2\,\mathring{k}_{\nu_s}\cdot\partial\big)^{L+1}\beta_s \hat{=} 0,\qquad \big(r^2\,\mathring{j}\cdot\partial\big)^{L+1}\beta_A \hat{=} 0.\tag{22}$$

Bjørhus form of (20) (the correction drives the incoming Weyl object to its CCM value, the Z4c analog of Ref. [3]):

$$U^-_{AB} - U^-_{AB}\big|_{\rm BC} \to 0,\qquad U^-_{AB}\big|_{\rm BC} = -\Big(\psi_0'\,\bar m_A \bar m_B + \bar\psi_0'\,m_A m_B\Big);\qquad \psi_0^{\rm CCE} \to 0 \implies \text{CPBC scheme of Ref. [1]}\quad \text{[V6]}.\tag{23}$$

Equations (21)–(22) are the order-L conditions of Ref. [1]; (22) replaces the Sommerfeld gauge placeholder of Ref. [3].

F. Sector orthogonality (nodes N4, N5)

For arbitrary TT data $h_{AB}(t, x_s)$ [V4]:

$$\mathcal{H} = \partial_i\partial_j h_{ij} - \partial^2 h^k{}_k = 0,\qquad \mathcal{M}_i = \partial_j \dot h_{ij} - \partial_i \dot h^j{}_j = 0,\qquad \gamma^{ij}\partial_t h_{ij} = 0,\qquad \tilde\Gamma^i = -\partial_j \tilde h^{ij} = 0;\tag{24}$$

negative controls: $h_{x_s y} = G(t, x_s) \Rightarrow \mathcal{M}_y = -\frac{1}{2}\,\partial_{x_s}\dot G \neq 0$; $h_{yy} = h_{zz} = G \Rightarrow \mathcal{H} = -2\,\partial^2_{x_s} G \neq 0$.

Hence the datum (20) sources neither (21) nor (22) at linear order: the three sectors of (20)–(22) commute.

On the boundary wave model of Ref. [1], $\big[\partial_t^2 - 2\mathring\beta\,\partial_t\partial_x - (1-\mathring\beta^2)\partial_x^2\big]u = 0$, plane-wave reflection [V5]:

$$R_{\rm Sommerfeld} = -\frac{\mathring\beta\,(1-\mathring\beta)}{(1+\mathring\beta)(2-\mathring\beta)} \neq 0,\qquad R_{(\partial_t + c_+\partial_x)} = 0\quad \text{(identically)}.\tag{25}$$

G. Frozen-coefficient boundary stability (node N7)

Laplace–Fourier ansatz $u = \tilde u\, e^{st+\lambda x}$, $\Re s > 0$ [V7]:

$$\lambda_+ = \frac{s}{1+\mathring\beta}\ \text{(admissible)},\qquad \lambda_- = -\frac{s}{1-\mathring\beta},\qquad D_L(s) = \Big(s + (1+\mathring\beta)\,\lambda_+\Big)^{L+1} = (2s)^{L+1},\qquad \frac{|D_L(s)|}{|s|^{L+1}} = 2^{L+1}.\tag{26}$$

D_L contains no datum symbol: the injection (20) modifies only the inhomogeneity, so the boundary-stability results of Ref. [1] (constraint sector: all L; gauge+metric sector: spherical reduction, asymptotically harmonic shift) transfer verbatim with $\|u\| \le C\,\|\psi_0'\text{-datum}\|$.

H. Numerical evidence

All runs: JAX float64, 4× NVIDIA A100-80GB (`CUDA_VISIBLE_DEVICES=0,1,2,3`), wall clock ≤ 23 s each (budget 600 s); outputs under `results/numerical/`.

Table I. Symbolic and sampling verification of (10)–(26). "Exact" = sympy identity on arbitrary functions or exact rationals (zero float error). Files: `n2_boost_check.txt`, `n3_exact_check.txt`, `n3_dictionary_check.txt`, `n1_varmap_check.txt`, `n4_cpbc_compat_check.txt`, `n5_gauge_check.txt`, `n6_composite_check.txt`, `n7_lf_sketch_check.txt`.

	verified statement	exact part	GPU sweep (4 devices)
V1	boost identity (13), factor $\sqrt{2}/2$	12/12 rational points	9.34×10^{-16} over 4,194,304 samples
V2a	EXACT identity (17)–(18), $(\sigma,\kappa)=(1,-1)$, $\bar\psi_4$ partner	generic Weyl, all 10 dof	—
V2b	linearized on-shell limit (corollary)	arbitrary F_1, F_2	0.0 over 4,194,304 modes (prior form)
V2c	Gauss–Codazzi realization (19), Schwarzschild	Codazzi sign +1 derived	E: 7.7×10^{-17}, B: 3.9×10^{-17}, identity 8.5
V3	maps (7), closure (10)	full-symbol identities	round trip 4.10×10^{-16}, gg^{-1} 8.88×10^{-16}
V4	orthogonality (24) + controls	identities and non-vanishing controls	TT residual 0.0 vs control 12.3
V5	reflection (25)	exact R algebra	$\lvert R_{\rm Som}\rvert \ge 0.023$ on $\mathring\beta \in [0.05, 0.6]$; $R=0$ ex
V6	10-mode count, $\psi_0 \to 0$ reduction, wire-through (exact targets)	structural	reduction 0.0; chain residual 1.47×10^{-16}
V7	Kreiss condition (26), $L=0..3$	$D_L=(2s)^{L+1}$ exact	ratio deviation 3.2×10^{-14} (4,194,304 sam

Table II. Dynamical model evolution (FD2+RK4, shift $\mathring{\beta} = 0.3$, outgoing Gaussian pulse onto the boundary; one configuration per GPU; `n8_model1d_test.txt`). The orientation of the discrete model gives the $\mathring{\beta} \to -\mathring{\beta}$ mirror of (25); the exact characteristic ratio at the boundary is $q_{\mathrm{in}}/q_{\mathrm{out}} = \mathring{\beta}/(2 + \mathring{\beta})$.

boundary condition	measured	exact	deviation
Sommerfeld, q-ratio	0.1305	0.1304	0.03%
Sommerfeld, R_u	0.2423	0.2422	0.04%
$(\partial_t + c_+\partial_x)$ [P1/CCM op., N=4097]	$R = 3.24 \times 10^{-15}$	0	machine precision
CCM injection (23), rel. L_2 vs exact pulse	1.09×10^{-4}	0	two-way transparent

[1] M. Ruiz, D. Hilditch, and S. Bernuzzi, Constraint preserving boundary conditions for the Z4c formulation of general relativity (2010), local source: `ref-paper/arxiv-1010.0523v2/`, arXiv:1010.0523 [gr-qc].

[2] J. Moxon, M. A. Scheel, and S. A. Teukolsky, Improved Cauchy-characteristic evolution system for high-precision numerical relativity waveforms (2020), local source: `ref-paper/arxiv-2007.01339/`, arXiv:2007.01339 [gr-qc].

[3] S. Ma *et al.*, Fully relativistic three-dimensional Cauchy-characteristic matching for physical degrees of freedom (2023), local source: `ref-paper/arxiv-2308.10361/`, arXiv:2308.10361 [gr-qc].